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“Tikkun” Viewed from the Perspective of Systems Philosophy

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Abstract

Understanding the idea of tikkun, as it appears in *tikkun olam* and *tikkun ha’yetser*, can be enriched by ideas from systems philosophy. From a systems perspective, “moral evil” – the flaws of individual human beings, which create the need for *tikkun ha’yetser* – is a special case of “natural evil” – the dangers to life and especially to humanity from the natural world and from societal moral evil, which create the need for *tikkun olam*. Natural evil is in turn a special case of “metaphysical evil” – flaws in *olam* (which include those of *yetser*) that are ubiquitous, which may be viewed from the religious perspective of Creation or from the secular perspective of scientific abstraction. Systems theory and philosophy offer an account of metaphysical evil: The fundamental flaw of everything in *olam* is finitude, the consequence of which is incompleteness and/or inconsistency. From a religious perspective this account is a “secular theodicy” which implies the need for *tikkun* in the Kabbalistic sense.

Key words

tikkun, *tikkun olam*, *tikkun ha’yetser*, systems theory, systems philosophy, systems analysis, theodicy, Lurianic Kabbalah, ontology of problems, incompleteness

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Introduction

Understanding the idea of *tikkun*, as it appears in *tikkun olam* – the fixing or repair of the world -- and *tikkun ha’yetsar* – the fixing or repair of character -- can be enriched by ideas from systems philosophy. This may appear to be an unusual claim, since systems philosophy is rooted in systems theory, and the idea of *tikkun* may seem distant from any connection to science. But systems theory is not ordinary science. It is really a research project aimed at constructing an “exact and scientific metaphysics” (ESM) (Bunge 1973), where “metaphysics” means general ideas applicable to many different phenomena, “scientific” means drawing from and contributing to scientific theories, and “exact” means being formulated mathematically, at least as a goal. In its metaphysical aspect, systems theory is philosophically very relevant to many religious themes, including the idea of *tikkun*.

To be specific, it is a systems theoretic idea that beyond some minimal complexity most systems are flawed, and many types of flaws can be explicitly and precisely described using graph theory, information theory, non-linear dynamics, game theory, and other systems formalisms (Zwick 1984). A side remark that must be inserted here: “systems theory” is not a singular theory, but actually an ensemble of theories, and also models and ideas, which can represent many different ways in which systems are flawed. A systems theoretic ESM thus offers an ontology of problems (Zwick 1995), a general account of imperfection, where a state of affairs is viewed as flawed or problematic or imperfect because of its impact on living systems and on human beings in particular. From a systems perspective, “moral evil” – the flaws of individual human beings – which creates the need for *tikkun ha’yetsar* -- is a special case of “natural evil” – the dangers to life and especially to humanity from the natural world and from moral evil at a societal level – which creates the need for *tikkun olam*. And natural evil is in turn a special case of “metaphysical evil” – flaws in *olam* and *yetsar* that are ubiquitous and that when viewed abstractly merit being called “metaphysical.” Systems theory offers a scientific account of metaphysical evil. From a religious perspective such an account is a secular theodicy (Zwick 2008). From a Jewish mystical perspective, this scientific account of metaphysical evil explains the need for *tikkun* in the Kabbalistic sense.

Genesis

Olam is that which encompasses all systems but, in a reverse synecdoche in which a whole also represents its parts, *olam* is also paradigmatic for individual systems. This is suggested by the fact that the origin and nature of *olam* in the Bible’s concept of Creation is surprisingly close to the scientific notion of “system.” Here I draw on Jon Levenson’s (1987) interpretation of Genesis. In this interpretation, Creation is not *ex nihilo*, something out of nothing. Rather, there was a primordial condition before Creation, namely *tohu va’vohu*, absolute disorder. Creation involves the imposition of order upon this primal disorder and simultaneously the generation of distinctions – between the waters above and the waters below, i.e., between heaven and earth,

and between land and the waters below. These ideas – order and distinction – are precisely the two core ideas in the notion of “system.” A system is characterized (i) by the presence of order as opposed to disorder and (ii) by the existence of a distinction between it and its environment. System formation is the arising of this order and this distinction. Creation in Genesis can be viewed as an account of system formation.

While the vanquishing of disorder in Creation is good, disorder is not simply always bad. The dyad of order and disorder can be expressed equivalently as the dyad of constraint and variety, and while order is the unmarked (privileged, favored) term in the order-disorder dyad, variety is arguably the unmarked term in the equivalent alternative dyad. Also, sometimes order is bad, and system dissolution good. The dispersal of the Tower of Babel project is an example. This said, order is still naturally associated with Being, and Being is good. As the Modern Orthodox Rabbi Irving (Yitz) Greenberg notes in his recently published magnum opus (*The Triumph of Life* 2024, 23), producing disorder by waste is forbidden by “...the Rabbinic interpretation of the ethical precept *bal tash’hit*, ‘do not destroy.’ It is forbidden to waste, to use up for no purpose or value, any object.” In systems theoretic terms, this *halakha* prohibits aiding, i.e., accelerating, the Second Law of Thermodynamics, which mandates the increase of disorder in isolated systems. The *halakha* does not actually forbid producing disorder in general; only destruction of *useful* order is forbidden. The matter is still more complex: Sometimes the joint presence of order and disorder can achieve results neither can achieve alone. Greenberg notes (16) that an “exact balance between order and disorder enables the emergence of life.”

What was created was, however, flawed. The primal condition was not totally vanquished by the order and distinction established by God’s decree. The entire Biblical story that follows attests to the imperfection of *olam*. One of the first indications of this imperfection occurs in the account of the Garden of Eden. Not all was perfect in this idyllic first human abode. There is a snake in the Garden. Adam and Eve yield to the temptation instigated by the snake and as a consequence are expelled from this slightly imperfect Garden into a much more imperfect *olam*. Cursed by being forced to live in a realm afflicted with natural evil (the pains and perils of childbirth, the difficult labor required to secure sustenance, etc.), specifically human forms of natural evil soon make their appearance. Cain slays Abel. Moral evil abounds, and God feels compelled to eradicate all life and start anew with Noah and his family and sample pairs of all living creatures. And so it went. If, anticipating Creation, there was concern about it among the angels, one angel might have reassured another that there was no need to worry about the soon to exist *olam*. After all, Creation would be the act of an all-powerful and all-good God. What could possibly go wrong? The answer, the Bible tells us, is everything, despite the fact that before resting on the Sabbath God observes that *olam* is good, and aspects of it are even very good. Genesis immediately poses the troubling question of theodicy, of how evil can exist in a world created by an all-powerful and good God.

Despite the Christian doctrine of “original sin,” what went wrong in Eden was *not* primarily due to human sin, since the actions of Adam and Eve were only a very mild form of moral evil. Why, after all, was there a snake in the Garden? Why was the Tree of Life so accessible? Why were Adam and Eve given natures that were so gullible and weak? As Rabbi David Cooper (2010) notes, “There are too many holes in the logic [of this story].” The imperfection of *olam* was not due to “original sin” but rather to the “sin of origin,” i.e., to the intrinsic and necessary imperfection that always accompanies system formation. Right at the outset in Genesis, the Bible poses the conundrum of an imperfect creation. *Olam* is good, but God does not say that it is perfect. It isn’t. The Midrash (Bereishit Rabbah 11:6) says, “Everything that came into being during the six days of Creation requires improvement—for example, the mustard seed needs to be sweetened ... also people need rectification.”

In fact, if one examines the notion of perfection closely, one sees that it is self-contradictory. If *olam* were perfect, it would have to be static, since any change would necessarily mar its perfection. But there is nothing perfect about stasis. Only perfecting, not perfection, is not self-contradictory. Perfecting is *tikkun*. Another way of seeing that perfection is inconceivable is to note that only God is perfect. If the created world were perfect as well, there would be two Gods, clearly impossible. To put this philosophically, *olam* is finite, and the root of all imperfections, problems, or flaws is finitude. To generalize (and modify) Gödel (1962): Beyond some minimal complexity all systems are incomplete and/or inconsistent. This generalization of Gödel is a succinct distillation of a systems theoretic ontology of problems.

Is it really plausible to see in Genesis a systems philosophical assertion that all (or most) systems are flawed? More generally is it really plausible to see philosophical messages in Biblical narratives? It is. As argued by Yoram Hazony (2012) the Bible can be considered to offer philosophical ideas conveyed not via rational and systematic exposition in the style of the Greeks but via the narration of stories that set out archetypal exemplars of general ideas. For example, Hazony points to passages in the Bible that suggest a notion of “truth” that supplements the traditional philosophical coherence and correspondence theories of truth with a third notion, namely a historical or pragmatic theory of truth, truth as what works, what has proven itself over time.

Lurianic Kabbalah

The Biblical account of the flawedness of Creation is radically deepened in Lurianic Kabbalah in its notion of the catastrophe and cataclysm of *shevirat ha’kaylim*, the breaking of the vessels, the inability of the vessels that were to contain the creative influx of divine power to bear this influx. As a consequence the vessels broke and sparks of divinity were scattered and trapped in the shards of these vessels in the created *olam*. This account is a theological myth about metaphysical evil. The generation of finitude from the infinite power of God cannot fail to

generate flaws, imperfections, problems. A consequence of the necessary flawedness of Creation is the necessary obligation of *tikkun*, thus tying the metaphysical and the ethical inextricably together. The responsibility for *tikkun* resides in life, more specifically in human life, acting as God’s partner, whose avant guard – according to the Judeo-centric Bible – is the Jewish people, not less flawed than any other aspect of Creation, as the Bible itself makes quite clear.

Creation is flawed by natural evil, or, expressed more generally, by metaphysical evil, so God’s partner, which is life, more specifically humanity, charged with *tikkun* of the created *olam* is necessarily also flawed by moral evil. So the task of *tikkun*, which is the task of Redemption, needs assistance in the form of inspiration and guidance. It receives this assistance in Revelation, and the linkage of Creation, Revelation, and Redemption is beautifully described in Franz Rosenzweig’s *The Star of Redemption* (1921). In *The Star*, Creation is the relation of God and World (*olam*). Revelation is the relation of God and Humanity – the love of God for individual human beings. Redemption is the relation of Humanity with World – the relation of *tikkun*. While Redemption is the charge of Humanity, it is not only World that is to be redeemed, but Humanity as well. And God too would be redeemed by an initially flawed Creation that has finally been redeemed.

That Redemption is the obligation and responsibility of life and more specifically of Humanity is expressed powerfully by Rosenzweig. Greenberg (2024) agrees and argues that the core value of Judaism is the protection and empowerment of life. This has two aspects: the goal of Creation is life, specifically human life: Individual human life is in the image of God. And it is life, specifically human life that is charged with the responsibility for the rectification of Creation, which reflects humanity’s partnership with God.

It is noted above that systems theory illuminates the ubiquitous flaws that afflict a wide variety of systems. It helps us understand that metaphysical evil is *lawful* and thus that *olam* needs *tikkun*. But systems theory is also supplemented by a worldly project that can be called systems analysis which is concerned with the methodologies of *tikkun* and focused on *l’ma’aseh*, on deed. It is a realization of modern science that the challenges faced by contemporary society are complex and not readily solved by simplistic or reductionist approaches. To cite one example of the contribution of the systems thought to *tikkun olam* in a literal sense: It was the systems community in the 1970s with its pioneering limits to growth studies (Forrester 1971, Meadows et al. 1972) that brought to the world’s attention the finitude of earth’s carrying capacity and the dangers of ecological collapse.

Rosenzweig insisted that ethical movements in modernity, in their secularity and even their atheism, play an essential role in bringing humanity closer to Redemption. Greenberg (7-8) concurs, arguing for “modern culture’s invitation to humans to take power and improve life

through secular activity, in response to the Divine call for partnership” and for “partnerships between religions and between religious and secular redemptive movements.” It might appear that Greenberg goes even further than Rosenzweig, in characterizing modernity as an era of the total *tsimtsum* of God, and the turning over the task of Redemption completely to humanity. Systems theory and systems analysis are important aspects of modern scientific culture that participate in this undertaking.

Tikkun Ha’yetser

Redemption of World, *tikkun olam*, requires as a prerequisite or concomitant the Redemption of Humanity. This means *tikkun ha’yetser*. We ourselves are flawed in our natures, our *yetser*s. The mildest way that this can be expressed is to say that we are incomplete. Morinis (2007) writes: The great Mussar teacher Rabbi Moshe Chaim Luzzatto discusses this notion in his book *Discerning Knowledge*: “The one stone on which the entire building rests is the concept that God wants each person to complete himself body and soul...” This teaching tells us that we are created incomplete so we can complete the work of our own creation. Our incompleteness manifests in the imperfections of our *middot*-- our strong negative emotions such as anger, pride, depression, etc., and the actions these emotions lead us to. Why express flaws of character in terms of the emotionally neutral word "incompleteness"? Because incompleteness and/or inconsistency, which are both due to finitude, are at the root of all flaws, problems, imperfections. Incompleteness (and/or inconsistency) of *olam* and of *yetser* are precisely how a systems theoretic “exact and scientific metaphysics” fundamentally characterizes the world.

A more direct and detailed formulation of the task of self-completion would speak of our *yetser tov* and our *yetser ra*. Completion is really *tikkun* of the latter, lower *yetser*. But note: Jewish thought says that we are born with a *yetser ra*, and only later develop our *yetser tov*, and also that our *yetser ra* is not really *ra*, not really bad, but in fact necessary for a fully human life. It is natural and only needs to be tamed and to serve our higher *yetser*. That our *yetser ra* is wholly natural is a clear statement that moral evil is a special case of natural evil. With the self as with the world, perfection is impossible, but perfecting is always possible, and obligatory.

The ontology of problems offered by systems theory has a lot to say about human imperfection, about the flaws of human understanding, of human character, and of human action, both at an individual level and at a collective level. It is not an accident that spiritual disciplines that guide people in inner work on *yetser* often make use of systems ideas to express the challenge of personal *tikkun* and point to ways by which this challenge might be met (Zwick 2010).

References

- Bunge, Mario (1973). *Method, Model, and Matter*. Boston: D. Reidel.
- Cooper, Rabbi David (2010). “The Mystical Garden of Eden.” Accessed 10/12/2024.
<https://www.rabbidavidcooper.com/cooper-print-index/2010/11/7/2364-the-mystical-garden-of-eden.html> .
- Forrester, Jay W. (1971). *World Dynamics*. Cambridge: Wright-Allen Press.
- Gödel, Kurt (1962). *On Formally Undecidable Propositions of Principia Mathematica and Related Systems*. Meltzer, B., transl., New York: Basic Books.
- Greenberg, Irving (Yitz) (2024). *The Triumph of Life: A Narrative Theology of Judaism*. Philadelphia: The Jewish Publication Society (Lincoln: University of Nebraska Press).
- Hazon, Yoram (2012). *The Philosophy of Hebrew Scripture*. Cambridge U.K.: Cambridge University Press.
- Levenson, Jon D. (1987). *Creation and the Persistence of Evil: The Jewish Drama of Divine Omnipotence*. New York: Harper and Row. Also Princeton: Princeton University Press.
- Meadows, Donella H.; Meadows, Dennis L.; Randers, Jorgen; and Behrens III, William W. (1972). *Limits to Growth*. New York: Universe Books.
- Morinis, Alan (2007). *Everyday Holiness: The Jewish Spiritual Path of Mussar*. Boston: Trumpeter.
- Rosenzweig, Franz (1921, translated 2005). *The Star of Redemption*. Galli, Barbara E., transl., Madison: University of Wisconsin Press.
- Zwick, Martin (1984). “Incompleteness, Negation, Hazard: On the Precariousness of Systems.” *Nature and System*, 6, pp. 33-42. <https://archives.pdx.edu/ds/psu/36224>
- Zwick, Martin (1995). “Towards an Ontology of Problems.” *Advances in Systems Science and Applications Special Issue I*, pp. 37-42. <https://archives.pdx.edu/ds/psu/36214>
- Zwick, Martin (2008). “A Conversation on Theodicy.” *Metanexus*, Jan 9.
<https://metanexus.net/conversation-theodicy/>
- Zwick, Martin (2010). “Personal Knowledge and the Inner Sciences.” In: Locker, Markus, ed., *Systems Theory and Theology: The Living Interplay between Science and Religion*, pp. 49-61. Eugene: Pickwick Publications.
<https://drive.google.com/file/d/1tIgGOV-Z47Dx4uktvSjKRMk3O7hsabB/view?usp=sharing>